



after five iterations on the Branin function, appendix B.3. The right plot shows the interval objective function values versus their radii, which is useful for identifying intervals to split with further evaluations.



Figure 7.13. The left plot shows the Lipschitz lower bounds constructed for the DIRECT intervals using the Lipschitz constant $\ell = 200$, and highlights one interval. The right plot shows how the minimum value for the lowerbound within the highlighted interval is the same as the y -intercept for a line of slope ℓ passing through that interval's $(r, f(\mathbf{c}))$ point.